# Questions: PMFs, PDFs, and CDFs

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**Summary**

A selection of questions to test your understanding of probability mass functions (PMFs), probability density functions (PDFs), and cumulative distribution functions (CDFs).

*Before attempting these questions it is highly recommended that you read Guide: PMFs, PDFs, and CDFs.*

## Q1

For each of the scenarios below, determine if the given distribution is a valid PMF and answer the following questions.

### 1.1.

Let $X$ be the random variable representing the result of rolling a biased four sided-die. The PMF of $X$ is given by:

| $x$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{10}$ | $\frac{1}{5}$ | $\frac{1}{2}$ | $\frac{1}{5}$ |

What is $P(X = 4)$?

### 1.2.

A discrete random variable $X$ has five possible outcomes $(1, 2, 3, 4,$ or $5)$, and the PMF is given by:

| $x$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|
| $P(X = x)$ | $0.25$ | $0.35$ | $0.05$ | $0.2$ | $0.1$ |

What is the probability of $X = 3$ or $X = 4$?

**1.3.**

A coin is tossed, where the probability of tails is $70$ and heads is $30$. Let $X$ represent the result of the coin toss. Complete the table below:

| $x$ | Heads | Tails |
|---|---|---|
| $P(X = x)$ | | |

**1.4.**

A discrete random variable $X$ has the possible outcomes $1, 2, 3, 4, 5, 6$, or $7$, with the following PMF:

| $x$ | $1$ | $2$ | $3$ | $4$ | $5$ | $6$ | $7$ |
|---|---|---|---|---|---|---|---|
| $P(X = x)$ | $0.1$ | $0.05$ | $0.05$ | $0.3$ | $0.25$ | $0.75$ | $0.35$ |

Is this a valid PMF? Justify your answer either way.

**1.5.**

A bag contains $5$ red, $3$ blue, and $2$ green sweets from a sweet shop. Let $X$ represent the color of a randomly picked sweet:

(a) What is the probability of picking a blue sweet?

(b) Construct the PMF for this scenario by completing the table:

| $x$ | Red | Blue | Green |
|---|---|---|---|
| $P(X = x)$ | | | |

**1.6.**

The PMF for a random variable $X$ is given as:

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | $p$ | $2p$ | $3p$ | $4p$ |

(a) For what value of $p$ is this a valid PMF?

(b) For this value of $p$, what is $P(X = 3)$?

## Q2

For each of the scenarios below, determine if the given distribution is a valid PDF and answer the following questions.

### 2.1.

Let $X$ be a continuous random variable on the interval $[0, 2]$ with the PDF:

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

What is the probability that $X$ lies between 1 and 2?

### 2.2.

Let $X$ be a continuous random variable with the PDF:

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the probability that $X$ lies between $0.5$ and $1$?

(b) What is $P(0.25 \leq X \leq 0.75)$?

### 2.3.

Let $X$ be a continuous random variable uniformly distributed between $3$ and $7$. The PDF is:

$$f(x) = \begin{cases} \frac{1}{4} & \text{if } 3 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases}$$

What is the probability that $X$ lies between $3$ and $6$?

### 2.4.

The PDF of a random variable $X$ is given by:

$$f(x) = \begin{cases} \frac{1}{9} & \text{if } 1 \leq x \leq 4 \\ \frac{1}{4} & \text{if } 5 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases}$$

Is this a valid PDF? Justify your answer either way.

**2.5.**

Consider the PDF:

$$f(x) = \begin{cases} kx^2 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) For what value of $k$ is this a valid PDF?

(b) For this value of $k$, what is $P(0.2 \leq x \leq 0.3)$?

**2.6.**

The PDF of $X$ is given by:

$$f(x) = \begin{cases} 4x & \text{if } 0 \leq x < 0.5, \\ 4 - 4x & \text{if } 0.5 \leq x < 0.75, \\ 0.5 & \text{if } 0.75 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Is this a valid PDF? Justify your answer either way.

## Q3

For each of the scenarios below, answer the following questions.

**3.1.**

In a scenario involving a discrete random variable, the following CDF is given:

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.3 | 0.5 | 1 |

(a) What is $F(3)$?

(b) What is $P(X > 2)$?

**3.2.**

For the random variable uniformly distributed on $[0, 2]$ as seen in Q2.2:

(a) Calculate the CDF at values $0.5$, $1$, and $2$.

(b) What is $F(3)$?

**3.3.**

For the PDF given in Q2.3:

(a) Calculate the CDF at points $x = 4$, $x = 5$, and $x = 6$.

(b) What is $P(X > 5)$?

**3.4.**

The CDF of $X$ for a scenario is given by:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.2 | 0.5 | 0.4 | 0.8 | 1 |

Is this a valid CDF? Justify your answer either way.

After attempting the questions above, please click this link to find the answers.

## Version history and licensing

v1.0: initial version created 12/24 by Sophie Chowgule as part of a University of St Andrews VIP project.

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